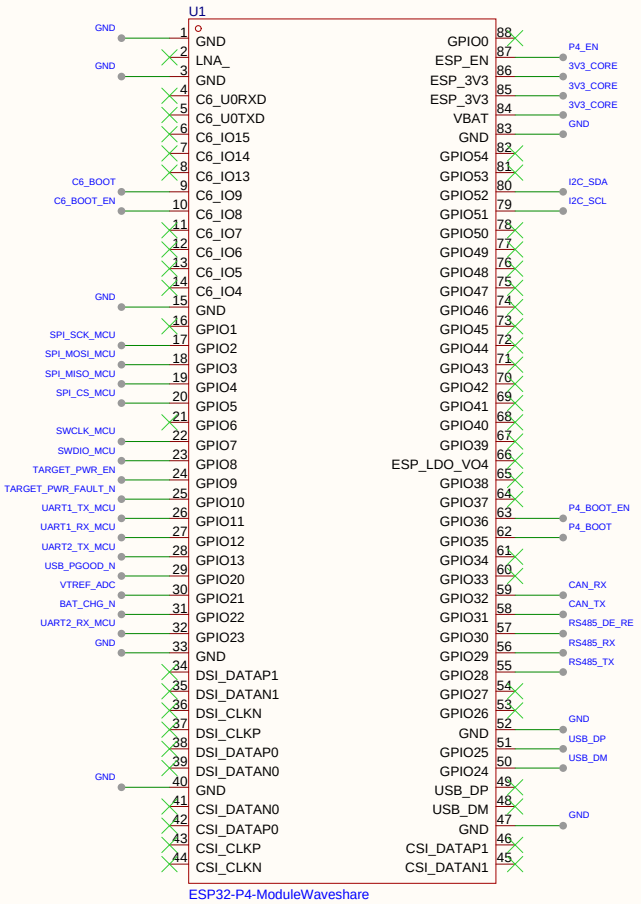
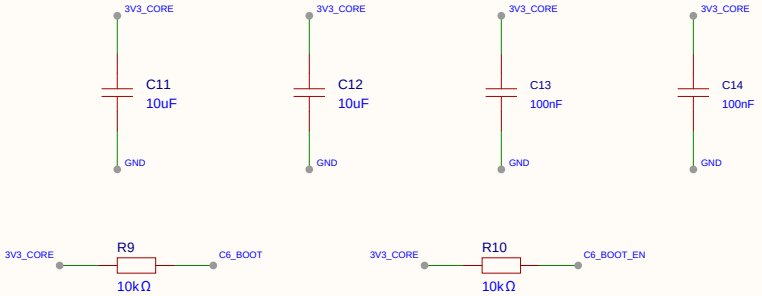
01 主控与电源 — 单节锂电 / USB充电 / P4 + C6

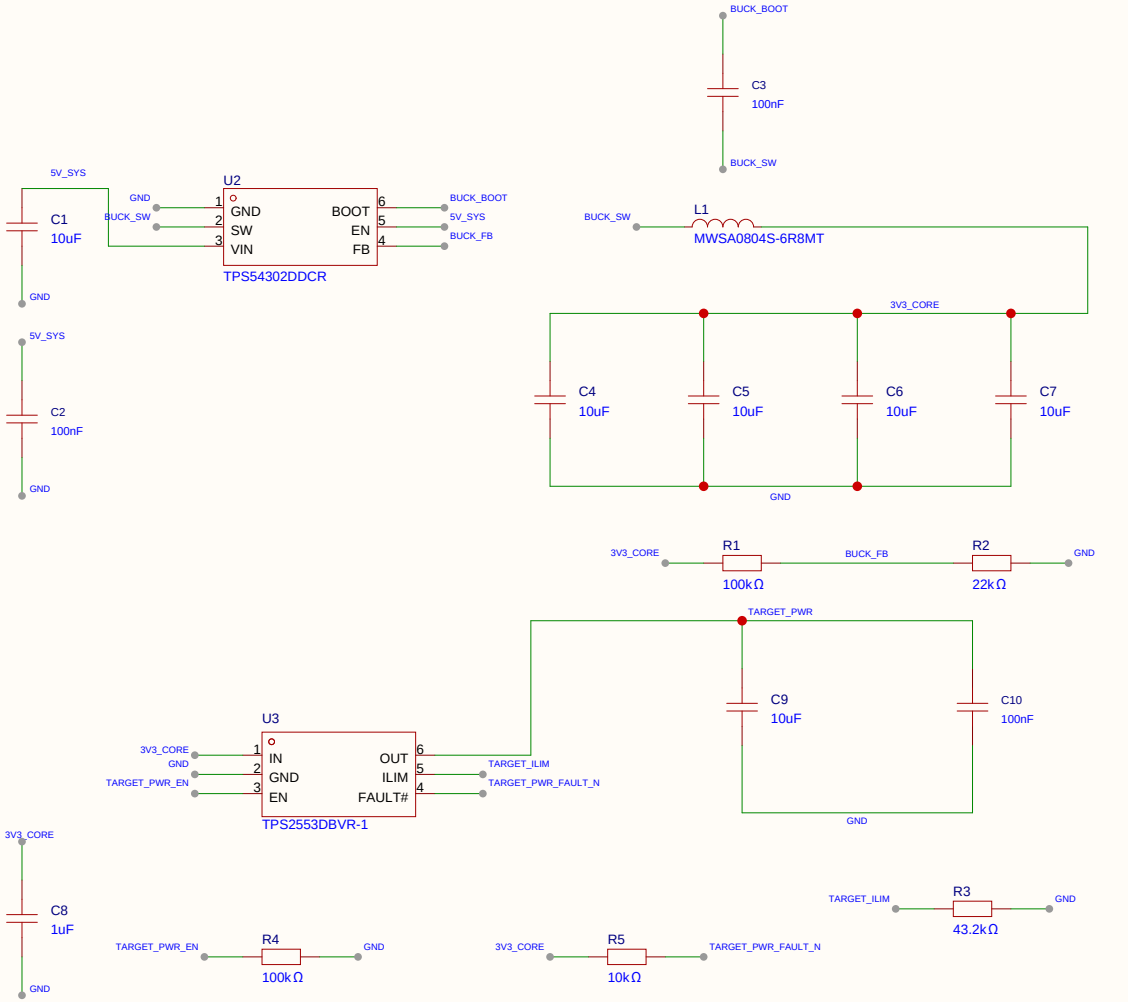
U1：Wi-Fi + BLE由模组内C6负责，P4本地执行SWD。



天线：模组IPEX → 外置2.4GHz胶贴/FPC；切换模组RF 0欧。

GPIO6/14~19/54内部占用；GPIO39~48默认1.8V域，不用于目标IO。

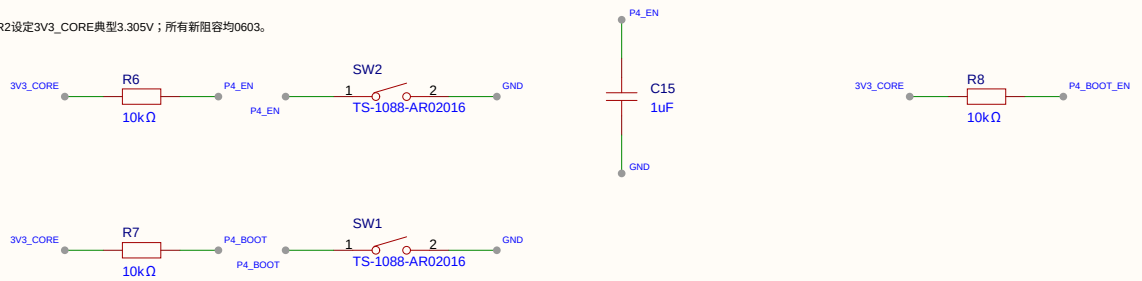




目标受控输出固定3.3V；U3默认关闭，过流锁断。

取消JP1；5V_SYS由第4页升压产生，另提供5V辅助输出。

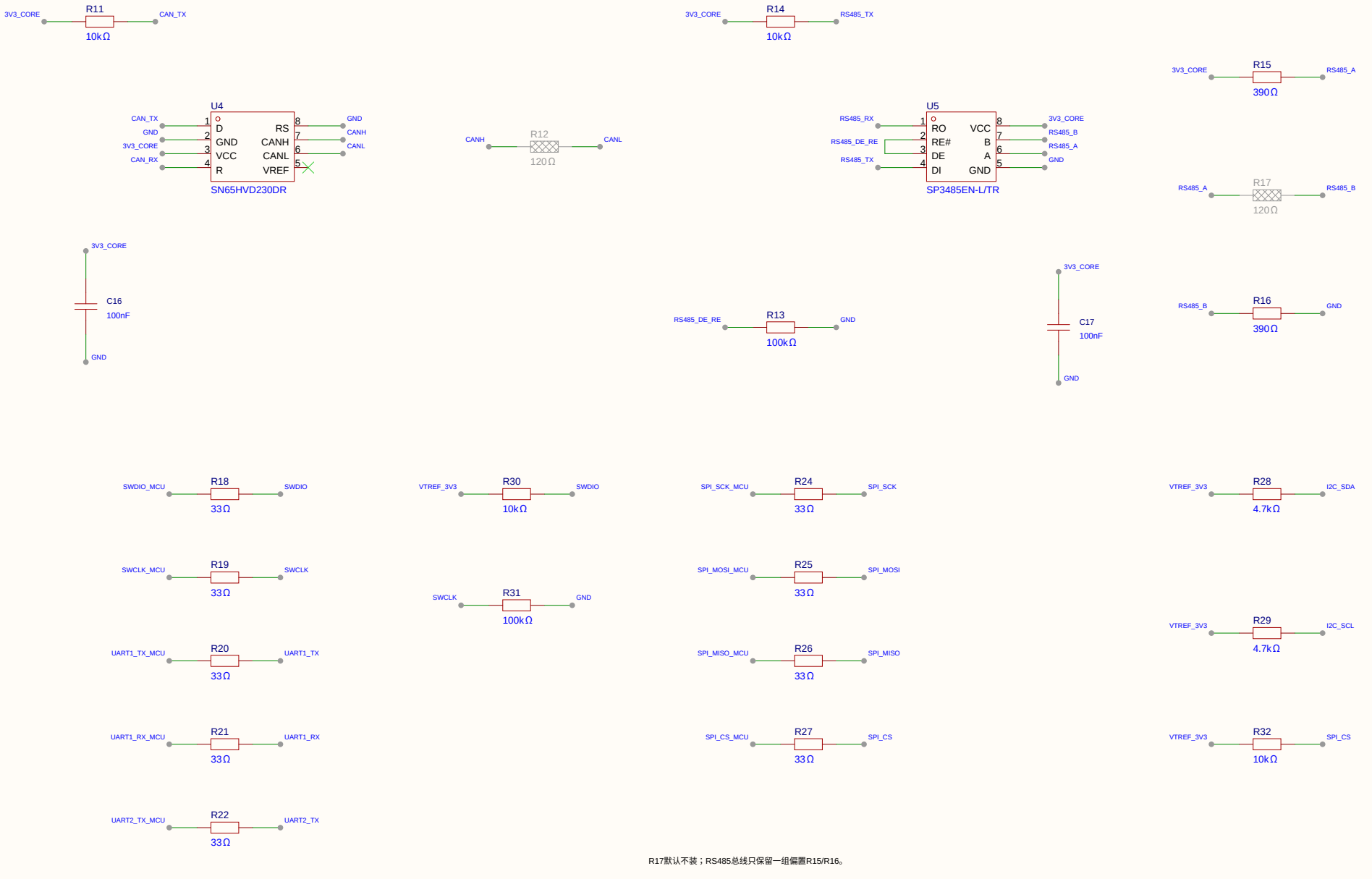
R1/R2设定3V3_CORE典型3.305V；所有新阻容均0603。



02 通信收发器与目标信号 — 仅3.3V逻辑

经典CAN：RS接地；R12默认不装，仅在总线端点装120欧。

半双工RS485：DE和/RE同控，默认接收。



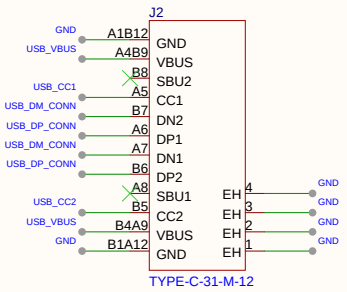
R17默认不装；RS485总线只保留一组偏置R15/R16。

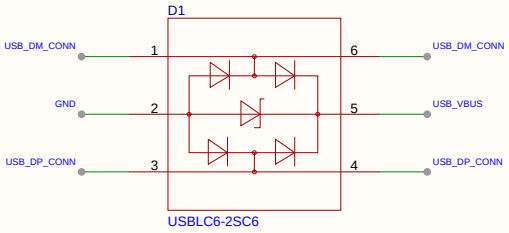
固件先检测VTREF，再驱动SWD/UART/SPI；掉电前全部高阻。

I2C与SWDIO/SPI_CS上拉到目标VTREF，禁止外接5V逻辑。

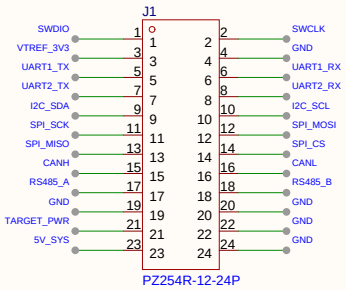
03 外部接口 — J1总IO / J2自身USB下载 / J3电池

USB Serial/JTAG：GPIO24=D-，GPIO25=D+。





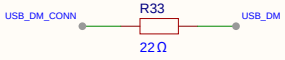
J1：2x12 / 2.54mm / 90度裸公头

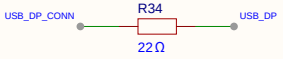


CC1、CC2分别接5.1k下拉；USB VBUS不与5V_SYS并联。

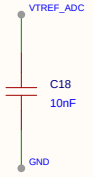




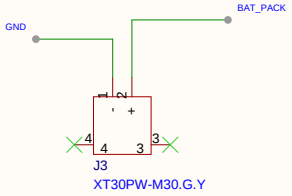


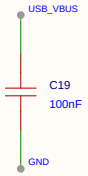






J3：1S 3.7V/3000mAh保护锂电池；Pin2正、Pin1负，禁止接5V。





J1-3接目标3.3V IO电源作VTREF；J1-21为受控3.3V供电。

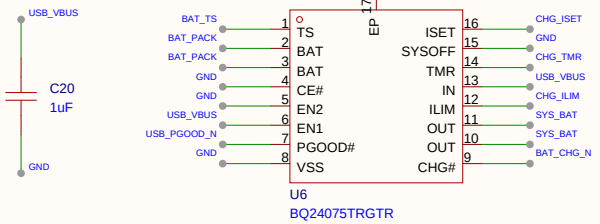
J1-23：原始5V辅助输出，绕过U3限流；不用时不接。

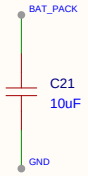
本版24Pin没有目标NRST，不能依靠复位脚进行连接/恢复。

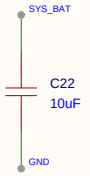
Type-C可给系统供电并给电池充电；SW3置ON后进行USB下载。

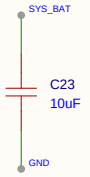
04 电池充电与升压 — USB供电 / 1S保护锂电 / 固定5V系统母线

J3：3.7V / 3000mAh / 605080扁平保护电池，4.2V充满；充放电接口共用。



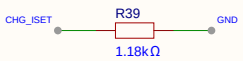


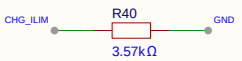


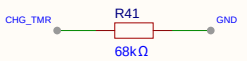


U6：USB总输入限流500mA；系统负载优先，实际充电小于500mA。

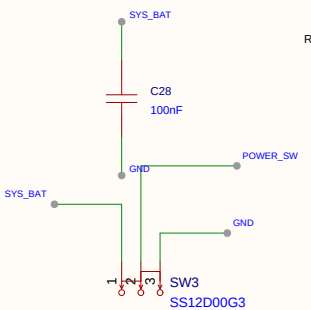
带载时充电电流自动减小，重载时电池补流；不保证满负载仍在充电。

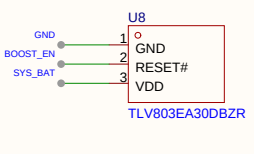


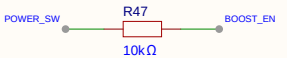


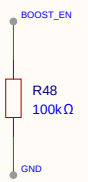


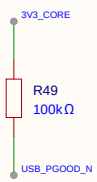
R39=1.18k，R41=68k；输入限流时安全计时按比例延长。

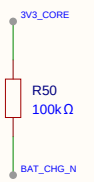








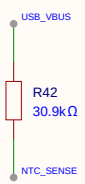


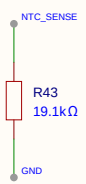


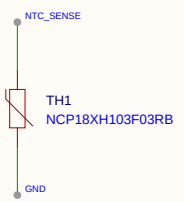
SW3：1-2=ON；2-3=OFF。关机切断升压，仍可USB充电。

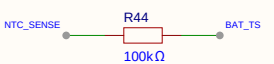
U8在SYS_BAT低于约3.08V时关断；非锁存，临界电量须停止烧录并充电。

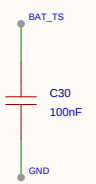
PGOOD/CHG接P4 GPIO20/22，只上拉到3V3_CORE。





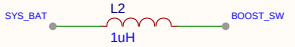


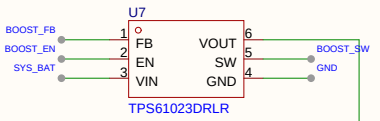


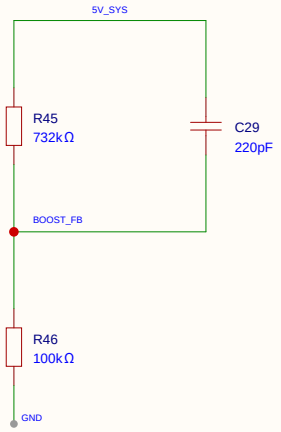


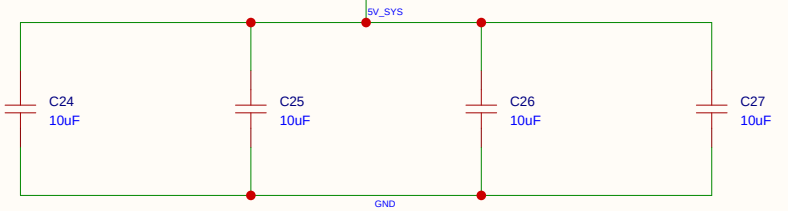
TH1必须在板背面贴合电池表面并绝缘，避开U6/U7/L2热源。NTC断线/短路会禁止充电。

NTC标称充电窗口约5~40℃；需实测热耦合和阈值。所有R/C/TH为0603。









USB输入仅5V；本板不请求USB-PD高压。